

Sports injuries: acute

with STUART WARDEN

After feeling discomfort in my hamstring after the first round last night and then again in the semi-final tonight, I was examined by the Chief Doctor of the National Championships and diagnosed with a grade I tear.

Usain Bolt, posted on Twitter 6 weeks before the Rio 2016 Olympic Games

Participation in sports and athletic endeavours has well-known health and wellness benefits. Unfortunately, participation is also associated with a risk of injury. Sporting injuries can occur during either competition or training and can affect any type of musculoskeletal connective tissue (Table 3.1). Injury to these tissues may be categorised as being either acute or overuse, based on the mechanism of injury and rapidity of symptom onset. The current chapter discusses acute injuries, whereas overuse injuries are described in the subsequent chapter (Chapter 4).

An acute injury refers to an injury that occurs during a single, identifiable traumatic event. They arise when the force applied to a tissue generates stresses and/or strains that are greater than the tissue can withstand. The net result is tissue failure which generates macroscopic damage and the rapid onset of symptoms, such as pain and loss of function. The severity of the symptoms depends on the tissue injured and the extent of the damage.

Forces applied to a tissue can be derived from either extrinsic (i.e. from outside the body, such as from a direct blow or collision with an external object) or intrinsic (i.e. from inside the body, such as from muscle contractile forces, or joint and tissue biomechanics) sources. Thus, acute injuries are frequently categorised as having extrinsic or intrinsic causes. Knowing the cause of acute injury has implications with regard to injury prevention as extrinsically generated forces may be modified by altering equipment, playing surface or sport rules, while intrinsically generated forces require an individual athlete's characteristics and capabilities to be considered (such as muscle strength, endurance and flexibility, motor control, joint range, biomechanics and proprioception). However, as an acute injury always results in intrinsic changes, an

individual athlete's characteristics and capabilities need to be considered in all instances of acute injury in order to facilitate return to participation and prevent re-injury.

When a force is applied to a tissue, the nature of the force being applied (such as the direction, magnitude and rate of loading) and the mechanical properties of the tissue in the direction of loading determine whether the tissue fails. The latter is determined by a combination of innate (non-modifiable) and environmental (modifiable) factors. Modifiable factors alter the safety factor between usual and injurious loads by either increasing or decreasing genetically endowed tissue strength. Modifiable factors include:

1. previous loading history and subsequent tissue adaptation
2. the presence and degree of any underlying microscopic tissue damage (overuse injury)
3. history of previous acute injury to the tissue and extent of mechanical strength recovery.

PATHOPHYSIOLOGY AND INITIAL MANAGEMENT

An acute injury initiates a common sequence of preliminary processes, irrespective of the specific tissue injured. Haematoma formation from damaged blood vessels gives way to an acute inflammatory response involving fluid exudation and phagocytosis. Fluid exudation contributes to oedema formation (i.e. swelling) and the delivery of white blood cells to the injured site, while phagocytosis aids the removal of damaged tissue and cellular debris. Swelling occurring within a joint is referred to as an 'effusion' and usually takes place slowly over the course of 12–24 hours. More rapidly occurring joint swelling (e.g. within the first 2–3 hours after injury)

Table 3.1 Classification of sporting injuries

Site	Acute injuries	Overuse injuries
Bone	Fracture (including growth plate) Periosteal contusion	Stress injury (including stress reaction and stress fracture) Osteitis, periostitis Apophysitis, enthesopathy Osteophyte/bone spur
Hyaline cartilage	Chondral/osteochondral injury	Chondropathy (e.g. softening, fibrillation, fissuring, chondromalacia)
Fibrocartilage	Acute tear (including meniscal, labral, and intra-articular and intervertebral discs) Herniation of nucleus pulposus from intervertebral disc	Degenerative tear Herniation of nucleus pulposus from intervertebral disc Chondropathy (e.g. softening, fibrillation, fissuring, chondromalacia)
Joint	Dislocation Subluxation	Synovitis Osteoarthritis/osteoarthritis Instability
Ligament	Sprain/tear (grades I–III)	Inflammation
Muscle	Strain/tear Contusion Cramp Acute compartment syndrome	Chronic compartment syndrome Delayed onset muscle soreness Focal tissue thickening/fibrosis
Tendon	Tear (partial or complete)	Tendinopathy (includes paratenonitis, tenosynovitis, tendinosis and tendinitis)
Fascia	Tear (partial or complete)	Fasciopathy (includes fasciitis and fasciosis)
Bursa	Traumatic bursitis	Bursitis/bursosis
Nerve	Neuropraxia	Entrapment Minor nerve injury/irritation Adverse neural tension
Fat pad	Bruise/contusion	Impingement/irritation
Skin	Laceration Abrasion Puncture wound	Blister Callus

indicates the effusion contains blood (i.e. haemarthrosis) and that an intra-articular structure has been damaged.

The acute inflammatory response to injury stimulates nociceptors to activate pain pathways. Pain is the body's way of protecting the injured site from further damage and leads to muscle inhibition and functional limitation. Muscle inhibition is also influenced by swelling. Reactive muscle spasm serves as an additional protective mechanism to further reduce potentially injurious loading, but also contributes to pain.

The acute inflammatory phase has the goals of protecting the damaged tissue from further injury, preparing the injured site for repair and stimulating the recruitment and activation of reparative cells. The inflammatory phase

is generally thought to persist for 48–72 hours; however, the duration within a particular individual can vary, and can be influenced by the tissue damaged, the severity of the damage and early management. Signs that acute inflammation is persistent include ongoing pain at rest and/or night, prolonged (>30 minutes) morning pain and stiffness and ongoing swelling that changes in volume according to recent activity.

Early management of acute injuries has historically centred on the PRICE acronym, which stands for protection, rest, ice, compression and elevation. Ice, compression and elevation are universally accepted, despite a lack of strong evidence for their efficacy,¹ and are introduced to reduce pain, decrease blood flow and

swelling and slow cellular metabolism to reduce risk of secondary injury (e.g. cell death due to hypoxia).

Similarly, there is general consensus for some degree of protection and rest immediately after injury; however, there is increasing acknowledgement that their duration should be limited due to the known detrimental effects of prolonged immobilisation and unloading of connective tissues. Instead, early loading is being increasingly advocated and shown as an important component of the early management of acute injuries. As a result, there are calls for the PRICE acronym to be changed to POLICE whereby 'rest' is replaced by 'optimal loading' (see Chapter 17).²

Optimal loading aims to take advantage of the responsiveness of musculoskeletal tissues to mechanical stimuli. It involves introducing progressive functional loading to stimulate connective tissue synthesis and promote healing, but without causing further damage. Optimal loading is principally guided by pain, with the induction of pain during loading indicating that tissue-level forces are too great for the current level of healing and that the load should be reduced to a pain-free level.

BONE Fracture

There is a large safety factor between the forces bone is exposed to during athletic activities and the forces required to generate a fracture. In the absence of a bone pathology that is causing generalised or localised bone fragility, large forces are required to fracture a bone. Such forces can result from direct trauma such as a blow or indirect trauma such as an awkward fall or twisting motion.

Traumatic fractures can be dichotomised as either closed (simple) or open (compound). The skin is intact over the fracture site in closed fractures, whereas in open fractures skin integrity is lost, often due to penetration from within by the fractured bone. The presence of a skin wound in open fractures that is continuous with the fracture site enables pathogens to enter and contribute to bone infection. Thus, individuals with an open fracture should be treated with prophylactic antibiotic therapy.

Within the dichotomy of open and closed fractures, fractures can be further classified according to their fracture pattern (Fig. 3.1), location in terms of both the affected bone and the location within that bone, amount and type of displacement of the fractured ends of the bone, and/or the specific name of the fracture. For instance, a Colles' fracture refers to a fracture of the distal radius with dorsal (posterior) and radial displacement of the distal fracture segment, whereas a Smith's fracture also refers to a fracture of the distal radius, but with ventral (anterior) displacement of the distal segment.

The immature skeleton presents some unique fractures because of its lower mineralisation and presence of unfused

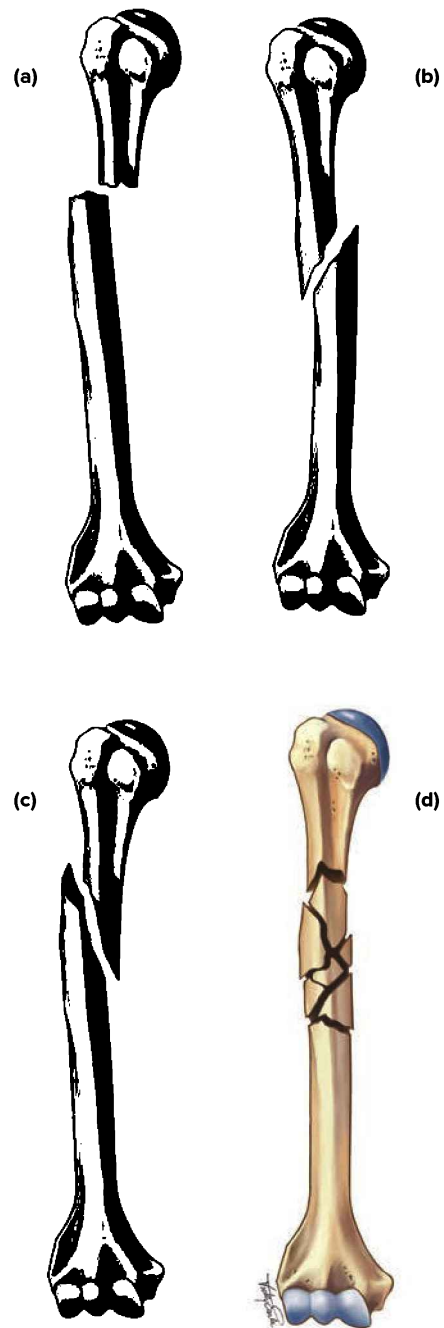


Figure 3.1 Types of fracture (a) Transverse (b) Oblique (c) Spiral (d) Comminuted

secondary ossification centres with interposed growth plates. The lower mineralisation of the immature skeleton makes it more flexible than the adult skeleton and, consequently, susceptible to buckling. When a sufficient bending force is applied to an immature bone the cortex on the side that the bone is bending toward is exposed

to compressive forces and buckles, while the cortex on the opposite side cracks open due to tensile forces. The resultant greenstick fracture is named according to how a green (i.e. fresh) stick buckles and cracks when bent. When the injurious force causes axial compressive loading of an immature bone as opposed to bending, the result can be the development of a torus or buckle fracture characterised by buckling and bulging of the cortex on all sides.

Growth plate fractures in children and adolescents present a particular problem and are discussed in detail in Chapter 43. The most common growth plate fracture is an avulsion fracture. Unfused secondary ossification centres represent an area of relative weakness in the young skeleton. They serve to develop bony prominences, such as tubercles and epicondyles, upon which tendons and ligaments attach. With an appropriate tensile pull from a tendon or ligament, a secondary ossification centre can be pulled away from the rest of the bone at its growth plate resulting in the development of an avulsion fracture. Avulsion fractures can also occur in the adult skeleton when extreme tensile forces at the insertion site of a tendon or ligament pull off a portion of bone.

The primary aim of fracture management is to allow the fracture to heal in the most anatomical position possible so that the mechanical function of the bone can be restored. If the fracture is displaced, it should be reduced to realign the bone fragments back to their anatomical position. A fracture that heals in an abnormal position (i.e. mal-united fracture) can lead to the long-term development of secondary complications, such as osteoarthritis.

Reduction is typically extremely painful and resisted by muscle spasm. Thus, it is usually performed under a short-acting anaesthetic, sedative or nerve block and with muscle relaxants. Once the fragments are reduced, the fracture needs to be held in place to permit fracture gap bridging. The method of stabilisation depends on the stability of the fracture. Stable fractures may be managed with a simple sling, cast or fracture brace, whereas more rigid stabilisation is required for unstable fractures that do not readily maintain their reduced position. Rigid stabilisation may consist of open reduction and internal fixation (ORIF) wherein the fracture site is opened surgically, reduced and fixed using metal plates, rods, screws, pins, and/or wires. ORIF has the advantage of allowing earlier introduction of forces across the fracture site. However, the trade-offs include:

- potential hardware loosening, failure and irritation
- opening the fracture site to the possibility for infection
- excessive stabilisation of the fracture leading to insufficient fragment micromotion and atrophic non-union
- peri-implant fracture resulting from regional osteoporosis due to stress shielding.

In addition to reducing a fracture and providing an environment that is permissive of bone regeneration, it is necessary to assess and monitor for other possible complications of fracture. Infection and mal-union have already been mentioned. Other acute complications that need consideration include acute compartment syndrome (discussed later), associated injuries (e.g. nerve, vessel), deep venous thrombosis/pulmonary embolism, and delayed or non-union.

Soft tissue injury, such as ligament or muscle damage, is often associated with a fracture and may cause more long-term problems than the fracture itself. Thus, it is important to address any concomitant soft tissue injury. Occasionally, deep venous thrombosis and pulmonary embolism may occur after a fracture, especially a lower limb fracture. This should be prevented by early movement and active muscle contraction.

Delayed or non-union of a fracture causes persistent pain and disability, and can be caused by excessive or insufficient fracture site stabilisation, presence of a complicated fracture type (e.g. open or highly comminuted fractures), insufficient or disrupted blood supply (e.g. due to abnormal regional anatomy or the presence of comorbid conditions such as diabetes) and lifestyle factors (e.g. smoking, excessive alcohol intake and poor nutrition), to name a few. Management options for delayed or non-united fractures include non-surgical treatments, such as low-intensity pulsed ultrasound and electromagnetic therapies (Chapter 16), as well as surgical treatments, such as bone graft or bone graft substitute, internal fixation and/or external fixation. In the future, biological compounds and small molecule pharmaceuticals may become available. Specific fractures that are common in athletes are discussed in Part B chapters.

Periosteal contusion

The periosteum is a dense, fibrous tissue that is firmly attached to the outer surface of bones and is both highly vascular and innervated. At subcutaneous skeletal sites (such as the medial surface of the tibia and iliac crest), the periosteum is susceptible to acute trauma from a direct blow with an external object, such as a ball, stick, opponent or playing surface. The blow damages periosteal blood vessels resulting in the development of a subperiosteal haematoma, otherwise known as a periosteal contusion. The confined space beneath the periosteum limits the spread of the haematoma, occasionally leading to the development of a palpable lump. The raised periosteum is tensed and inflamed/irritated, stimulating its plentiful nociceptive nerve endings to cause considerable pain, especially upon palpation and on contraction of muscles attaching to the injured region.

A periosteal contusion at the iliac crest, often referred to as a 'hip pointer' injury, can be extremely painful because of involvement of the cluneal nerve which runs along the iliac crest. Management of periosteal contusions focuses initially on minimising the extent of the haematoma using conventional first aid approaches followed by a gradual return to activity. Protective equipment (such as shin guards and padding) should be considered for future prevention.

HYALINE CARTILAGE

Chondral and osteochondral injuries

Hyaline, or articular, cartilage lines the articular surface of bone regions forming synovial joints. It provides a smooth, lubricated surface allowing for low-friction gliding and possesses unique viscoelastic properties which facilitate the absorption and distribution of loads to the underlying subchondral bone. In fulfilling these roles, hyaline cartilage is principally exposed to compressive loads. Acute joint subluxation or dislocation (discussed below) and acute ligament sprain or rupture (discussed later) can lead to locally high compressive forces and the superimposition of shearing forces due to excessive joint translation. The net result can be development of an acute chondral (cartilage), as shown in Figure 3.2a and b, or osteochondral injury, with the latter involving both the articular cartilage and underlying (subchondral) bone (Fig. 3.2c).

Articular cartilage is radiolucent and therefore most chondral injuries are not visible on conventional radiology. As a result, these injuries have historically been underdiagnosed. With the advent of magnetic resonance imaging and direct visualisation via arthroscopy, it has become clear that chondral injuries are far more common than previously realised. Clinicians must maintain a high index of suspicion for chondral involvement if an apparently 'simple joint sprain' remains painful and swollen for longer than expected, despite the presence of normal X-rays. Common sites for chondral and osteochondral injuries are the femoral condyles, superior articular surface of the talus, capitellum of the humerus and patella.

Chondral injuries are a concern due to their poor healing and their contribution to the development of premature osteoarthritis. Articular cartilage has limited regenerative and repair capacity due to its avascular nature. As larger lesions or defects have lower probability of healing, classification systems have been developed to quantify the severity of chondral damage and guide prognosis. Chondral lesion severity is determined by the depth the injury extends towards the underlying bone and the size of cartilage affected, whereas osteochondral lesion severity is determined by the continuity of the overlying cartilage

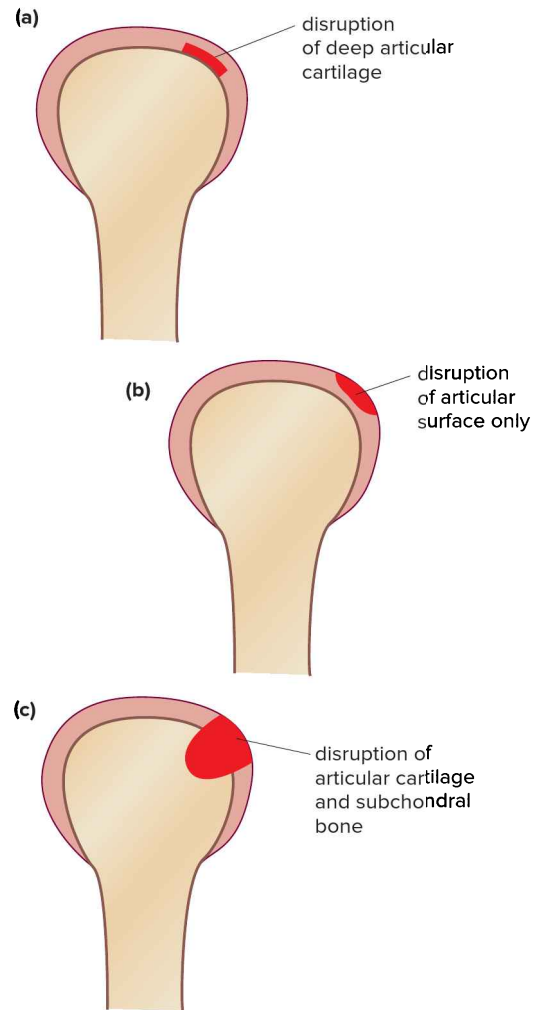


Figure 3.2 The three types of articular cartilage injury

and stability of the lesion. Scales generally extend from 0 (normal) to a high score of 4 (exposed bone for chondral injuries; fragment displacement for osteochondral lesions), with higher scores indicating greater damage and a worse prognosis.

Chondral and osteochondral injuries in young active individuals represent a therapeutic challenge. The goal of treatment is to restore the structural integrity and function of the affected surface so that it can withstand the significant stresses associated with athletic endeavours. The hope is to permit a return to participation while minimising cartilage degeneration and the need for early arthroplasty (joint replacement). Treatment options for chondral injuries range from palliative techniques (such as chondroplasty to smooth loose edges of damaged cartilage and arthroscopic washout to remove debris) to techniques aimed at stimulating

cartilage repair or restoration. Reparative techniques include bone marrow stimulation techniques such as abrasion arthroplasty, drilling and microfracture, whereas restorative approaches include autologous osteochondral mosaicplasty, osteochondral allograft transplantation, and chondrocyte and mesenchymal stem cell-based therapies wherein cells are implanted into the cartilage lesion and covered with a periosteal or collagen-membrane cover. Treatment choice depends upon the size of the chondral or osteochondral lesion, with great debate continuing as to the most effective approach. Ultimately, return to sport depends on a range of factors, including athlete age, duration of symptoms, level of play, repair tissue morphology, lesion size, type and location, and the number of surgeries and concomitant procedures.³

FIBROCARTILAGE

Acute tear

Fibrocartilage consists of a mixture of fibrous and cartilaginous tissue in varying proportions, which provide it with both toughness and elasticity. Fibrocartilage forms a range of structures in different joints with varying functions, but roles generally include enhancing joint stability, contributing to shock absorption and distribution and promoting joint lubrication. Example fibrocartilage structures include the knee menisci, glenoid and acetabular labrums, triangular fibrocartilage complex (TFCC) at the radio-carpal joint, volar plates of the digits, and articular discs within the acromioclavicular and sternoclavicular joints.

Given the predominantly mechanical roles of fibrocartilage structures in enhancing joint congruency and distributing stresses, it is not surprising that they are at risk of acute injury when excessive forces are introduced. For instance, the knee menisci are at risk when rotation is superimposed onto a flexed and loaded knee, whereas a fall onto an outstretched hand can acutely injure the TFCC at the radio-carpal joint. Joint fibrocartilage structures can be injured in isolation; however, simultaneous injury with associated joint structures (e.g. ligaments and joint capsule) is common due to their close anatomical and mechanical relationships. For instance, it is common for a knee meniscus and the anterior cruciate ligament to be simultaneously injured as they share a common mechanism of injury (i.e. rotating on a flexed and loaded knee). Similarly, during shoulder dislocation it is common for the anterior glenoid labrum to be simultaneously injured with the glenohumeral ligaments and joint capsule due to their anatomical connections and forces being introduced.

Signs and symptoms of acute fibrocartilage injury are region- and injury-specific, but generally include pain and

swelling in the affected joint combined with joint clicking, catching or locking. The latter joint events are often delayed in their presentation, occurring once the initial acute symptoms have subsided and normal use of the joint has recommenced. Joint clicking, catching or locking can contribute to reflex muscle inhibition and the sensation of the joint 'giving way' during load bearing.

Management of acute fibrocartilage injuries depends on the type, size and location of damage. Management options are dichotomised into conservative and surgical, with surgical options including repair, removal and replacement of the damaged fibrocartilaginous structure. Surgery is an option for individuals who have failed to respond to conservative management, but is also frequently used as a first-line management approach, particularly for injuries in regions that have limited healing potential. The latter regions include the relatively avascular zones within the inner two-thirds of the menisci and the central region of the TFCC.

Herniation of nucleus pulposus from intervertebral disc

Herniation of the nucleus pulposus in athletes most commonly occurs secondary to damage accumulation within the disc in response to repetitive flexion and/or rotational loading. Thus, the vast majority of disc herniations are overuse injuries, despite the onset of symptoms often occurring relatively spontaneously and in response to an apparently trivial loading event. However, it is possible to acutely prolapse or herniate an otherwise healthy intervertebral disc with sufficient compressive and flexion force. This can occur during contact sports when the spine is axially loaded while in a flexed position, such as when driven into the ground when being tackled. The disc may protrude or herniate radially beyond the usual margins of the annulus fibrosis or herniate through the vertebral body endplate into the vertebral body to form a Schmorl's node.

JOINT

Dislocation/subluxation

Joint stability depends on the interaction between the passive, active and neural subsystems. Muscles and tendons combine to form the active subsystem, which is controlled by the neural subsystem, to provide dynamic joint stability. When the force applied to a joint exceeds the capabilities of the active subsystem or when the active and/or neural subsystems are compromised, loads are transferred to the passive subsystem. The passive subsystem consists of non-contractile connective tissues, which includes the bony anatomy, joint capsule, ligaments and fibrocartilage joint structures (e.g. labrum, volar plates, menisci).

Excessive load placed on the passive subsystem can cause the bones forming a joint to abnormally translate or luxate relative to one another. When the bones are forced to completely separate so that the articulating surfaces are no longer in contact it is referred to as dislocation (Fig. 3.3a). Subluxation (partial dislocation) refers to when the bones shift relative to one another, but the articulating surfaces remain partially in contact (Fig. 3.3b).

Luxation of a joint invariably results in damage to passive subsystem structures. The structures damaged and extent of damage depends on the direction and magnitude of the luxation force and the inherent stability of the joint created by the passive subsystem. The hip joint is inherently stable because the large ball shape of the femoral head is well encased in a reciprocating socket-shaped acetabulum and the bones are supported by a strong joint capsule that is reinforced by robust ligaments. Thus, considerably more force is required to luxate the hip joint compared to a less stable joint, such as the shoulder.

The shoulder lacks inherent stability because the large humeral head outsizes the small, shallow glenoid fossa and the joint possesses a thin, loose capsule that is minimally supported by ligaments. The heightened force required to luxate inherently stable joints (such as the hip, elbow, ankle and subtalar joints) means that subluxation or dislocation of these joints is more likely to be associated with damage beyond that to the joint capsule and ligaments, including fractures and damage to cartilage, vessels and nerves.

A dislocated joint is readily identifiable by gross deformity with complete loss of joint function. The individual presents with significant, acute pain and often has the afflicted limb cradled or held. Following a neurovascular screen to assess for nerve and blood vessel damage, the dislocated joint should be reduced as quickly as possible. In most cases, this can be performed by applying gradual and controlled

distraction of the joint while simultaneously moving the joint through a passive range of motion. When distraction is unable to overcome the opposing muscle spasm and the joint does not readily reduce, use of an injected muscle relaxant or general anaesthetic may be required.

After reduction, all dislocated joints should be X-rayed for the presence of an accompanying fracture and the joint should be protected to allow the joint capsule and ligaments time to heal. However, early protected mobilisation is encouraged to promote functional healing, and training of the active and neural subsystems should commence as early as possible and progress to functional activities so that these subsystems may compensate for the reduced stability afforded by the damage to the passive subsystem. Unfortunately, in some joints (such as the shoulder), active and neural subsystem training is often insufficient to prevent re-dislocation or chronic subluxation, necessitating eventual surgical reconstruction of the damaged capsule and ligaments.

LIGAMENT

Sprain/tear

Ligaments typically span joints to connect articulating bones, and present as either discrete extra-articular structures or thickenings of the joint capsule. Their structure consisting of tightly packed bundles of collagen arranged nearly in parallel along the longitudinal axis of the ligament is designed to resist tensile loads. Ligaments function to provide passive stability and guide what directions of motion are available at a joint. When force is applied to a joint that attempts to move the bones in a direction they are not designed to move, passive tension rises in ligaments on the side of the joint being gapped open. When the load is sufficient, and dynamic joint stabilisation afforded by the active and neural subsystems is insufficient, collagen fibres within ligaments begin to yield resulting in an acute ligament injury (commonly referred to as a 'sprain').

Ligament sprains range in severity from mild injuries involving tearing of only a few fibres to complete tears of the ligament where ligament continuity and its stabilising role is completely lost. Based on the number of fibres torn and the subsequent degree of joint instability, ligament sprains are classified into three grades (Fig. 3.4), each representing an increase in injury severity (Table 3.2).

Management of acute ligament sprains is summarised in Figure 3.5. Initial management consists of first aid techniques to minimise bleeding and swelling (Chapter 17). For grade I and II sprains, subsequent treatment aims to promote tissue healing, prevent joint stiffness, protect against further damage and strengthen muscle to provide dynamic joint stability. Return to sport usually takes place before tissue level healing is complete, with healing of collagen in a partial

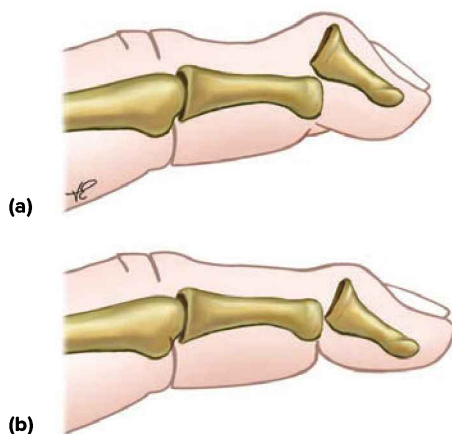


Figure 3.3 Acute joint injury (a) Dislocation (b) Subluxation

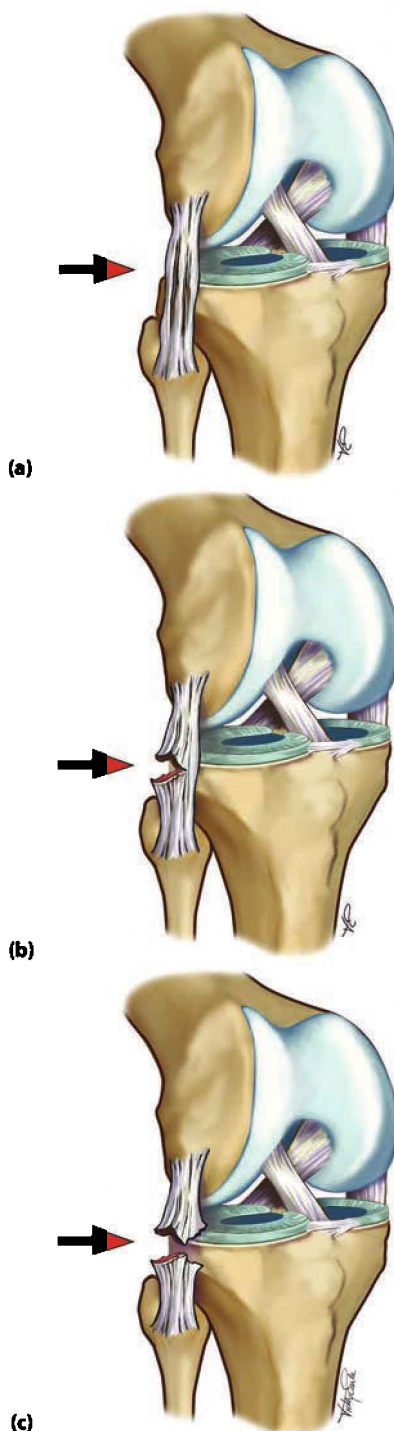


Figure 3.4 Ligament sprains (a) Grade I (b) Grade II (c) Grade III

ligament tear taking several months.^{4,5} Earlier return to sport is facilitated by the use of bracing or taping to help protect against re-injury. Rehabilitation (especially neuromuscular training) should continue in some form following return

Table 3.2 Classification of ligament injuries

Grade	Pathology	Clinical findings
I	Disruption of some collagen fibres	• Localised tenderness on palpation • Minimal inflammation (swelling) • Normal range and end feel on ligament stress test, but test may be painful • Little functional deficit
II	Disruption of considerable proportion of collagen fibres	• Significant tenderness on palpation • Can be considerable inflammation (swelling) involving the whole joint (effusion) • Increased joint play on ligament stress test (increased ligament laxity), but a definite end point is present • Moderate functional deficit
III	Complete disruption of all collagen fibres	• Audible 'pop' may have been heard at the time of injury • Often immediately painful, but may become pain-free a short time after the injurious event due to simultaneous disruption of nociceptors (pain returns thereafter due to effusion and inflammatory mediators activating alternative nociceptors) • Can be considerable swelling which may occur rapidly due to bleeding into the joint (haemarthrosis) • Significantly increased joint play on ligament stress test with no discernible end point • Significant functional deficit

to sport as injury risk is heightened, and individuals often have ongoing objective mechanical laxity and subjective instability. For instance, as much as one-third of individuals report ongoing pain and subjective instability one year following acute ankle sprain, while up to a third report at least one re-sprain within a period of 3 years.⁶

The treatment of a grade III sprain may be either conservative or surgical. For example, the torn medial collateral ligament of the knee and the torn lateral

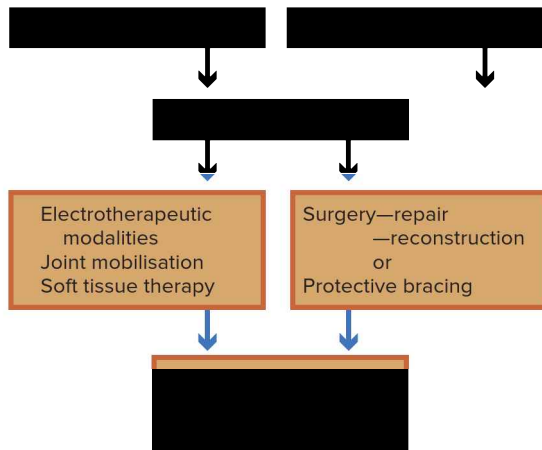


Figure 3.5 Management of acute ligament sprains

ligament of the ankle may be treated conservatively with full or partial immobilisation. Alternatively, the two ends of a torn ligament can be reattached surgically and the joint then fully or partially immobilised for approximately 6 weeks. In certain instances (e.g. anterior cruciate ligament rupture), torn ligament tissue is not amenable to primary repair and surgical ligament reconstruction may be required (Chapter 35).

During the past few decades, there have been mounting efforts to develop tissue engineering strategies that encourage ligament regeneration (as opposed to repair) so that the final product matches that of native ligament. Strategies have included the use of growth factors, cell-based therapies, gene transfer and therapy, and the use of scaffolding materials of varying origin.⁷ While these approaches have shown promise in preclinical studies, translation to clinical populations remains lacking and is required before clinical utility.

MUSCLE

Muscle injuries are among the most common injuries in sports, accounting for up to half of all sport-related injuries at the elite level. They result from either intrinsic or extrinsic causes, with the former contributing to muscle strains/tears and the latter resulting in contusions.

Strain/tear

A muscle is strained or torn when excessive tensile and/or shear forces within the muscle cause muscle fibres and their surrounding connective tissue to fail. Muscles consist of a hierarchy of active contractile elements and passive non-contractile connective tissue. The actin and myosin machinery responsible for producing active contractile forces form myofibrils which are housed in elongated, rod-shaped cells called muscle fibres. The cell membrane of the

muscle fibre (i.e. sarcolemma) attaches to a basal membrane surrounded by a connective tissue sheath called the endomysium. Groups of muscle fibres bundle together to form fascicles encased in a connective tissue sheath called the perimysium. The fascicles in turn combine to form the muscle which is surrounded by a final dense sheath of connective tissue called the epimysium.

During muscle contraction, actin and myosin form cross-bridges to generate force, which is transmitted either via the surrounding connective tissue or directly to an attached tendon and subsequently the skeleton, to produce motion. In muscle strains, excessive tensile force results in muscle fibre failure, which ruptures its cell membrane and the basal membrane to which the membrane attaches, damages the surrounding connective tissue sheaths and tears blood vessels contained within the connective tissue sheaths. Depending on the severity of the injury, the net result can be pain on active contraction and passive stretch of the muscle, a reduction in strength due to pain-inhibition, decreased range of motion due to muscle spasm and loss of function.

Muscle strains most commonly occur within the hamstring, quadriceps and gastrocnemius muscles, as these muscles are exposed to the highest amounts of total tension. Total tension is the sum of active tension generated by muscle fibre contractile forces and passive tension due to stretch on the connective tissue components. Active tension can be high in the hamstring, quadriceps and gastrocnemius muscles as these muscles often contract eccentrically and they contain a high proportion of type-II (fast-twitch) muscle fibres—both factors associated with the generation of greater active tension. Passive tension is also often high in these muscles as they are biarthrodial (cross two joints) and are often required to contract while on passive stretch over more than one joint.

To guide prognosis, muscle strains have historically been graded using a three-tiered system, with increasing grade suggesting a greater severity of injury requiring longer recovery.⁸ A minor strain causing localised pain but no or minimal loss of strength is referred to as a grade I injury and suggests the involvement of a small number of muscle fibres. A grade II strain involves a greater number of muscle fibres and is clinically evident by greater pain, swelling and loss of strength, while a complete tear of the muscle is classified as a grade III injury. In addition to these three grades, a grade 0 injury has been described wherein an athlete presents with a clinical syndrome of muscle abnormality, but without imaging evidence of pathology.

Grading systems provide an indication of the severity or extent of muscle damage; however, the variable recovery time for individuals with similarly graded injuries suggest that other factors influence prognosis. With advances in magnetic resonance imaging and ultrasonography, it

is now possible to also classify or categorise injuries in terms of their anatomical location and the type/s of tissue involved. The role of imaging in predicting recovery time from muscle strains remains unclear.⁹ However, there is increasing recognition of the benefits of combining clinical and radiological findings, which has resulted in the generation of a number of 'new' muscle injury grading and classification systems (Table 3.3 and Fig. 3.6).^{8, 10-12} These systems have yet to be convincingly validated,¹³ but may represent a new era in understanding and establishing prognosis for muscle strains.

Common to the new muscle strain grading systems is the recognition that strains involving tendinous components of the muscle have a worse prognosis. Muscle strains most commonly occur at or near the musculotendinous junction, which appears as an area of relative weakness in skeletally mature individuals. The traditional view of the muscle-tendon unit involved a relatively distinct separation between the muscle belly and the free tendon at the end via which active contractile forces are transmitted to the bone. However, advanced imaging and anatomical dissection studies have clearly shown that tendinous structures extend

deeply into the muscle belly and that some muscles also have isolated central tendinous structures within the muscle belly itself.

It is well established that muscle strains involving concomitant damage to the free tendon (e.g. proximal hamstring tendon) require a longer recovery time than strains isolated to the musculoskeletal junction. However, the presence of and damage to intramuscular and central tendons may explain why some individuals with historically labelled 'muscle belly strains' have a slower than expected recovery. The presence of intramuscular and central tendinous structures means that musculotendinous junctions occur throughout a greater length of a muscle and that 'muscle belly strains' still represent an injury occurring at one or more of these junctions. As tendinous structures have a slower healing rate than muscle tissue, a muscle strain that also involves damage to intramuscular or central tendinous structures has the potential to delay healing and return to play.¹³ Indeed, a recent cohort study reported that strains of the biceps femoris muscle with and without central tendon disruption required contrasting median recovery times of 72 and 21 days, respectively.¹⁴

Table 3.3 Comparison of classification systems for acute muscle injuries

O'Donoghue, 1962 ⁸	Chan et al., 2012 ¹⁰	Müller-Wohlfahrt et al., 2013 ¹¹	Pollock et al., 2014 ¹²
1. Mild No appreciable tissue tearing, no loss of function or strength, only a low-grade inflammatory response	1. Proximal musculotendinous junction	A. Indirect muscle disorder/injury <i>Functional muscle disorder</i> Type 1: Overexertion-related muscle disorder	0a.Focal neuromuscular injury with normal MRI
2. Moderate Tissue damage, strength of the musculotendinous unit reduced, some residual function	2. Muscle A. Proximal B. Middle C. Distal a. Intramuscular b. Myofascial c. Myofascial/perifascial d. Myotendinous e. Combined	1A: Fatigue-induced muscle disorder 1B: Delayed onset muscle soreness (DOMS) Type 2: Neuromuscular muscle disorder 2A: Spine-related neuromuscular 2B: Muscle-related neuromuscular	0b.Generalised muscle soreness (DOMS) with normal MRI
3. Severe Complete tear of musculotendinous unit, complete loss of function	3. Distal musculotendinous junction	<i>Structural muscle injury</i> Type 3: Partial muscle tear 3A: Minor partial muscle tear 3B: Moderate partial muscle tear Type 4: (Sub)total tear Subtotal or complete muscle tear Tendinous avulsion B. Direct muscle injury <i>Contusion</i> <i>Laceration</i>	1. Small injuries (tears) 2. Moderate injuries (tears) 3. Extensive tears 4. Complete tears a. Myofascial injury b. Within muscle usually at musculotendinous junction c. Extends into tendon

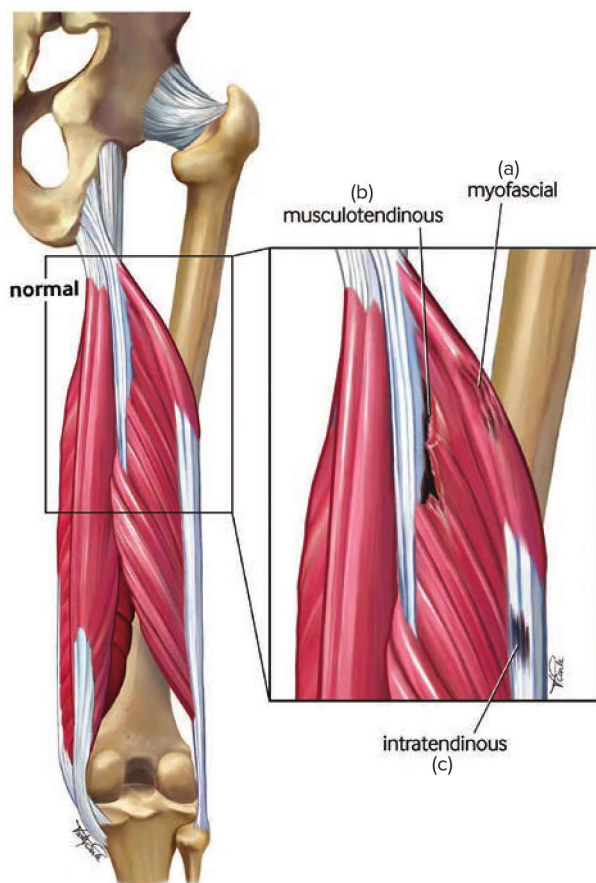


Figure 3.6 Letter classification based on the anatomical site of muscle injury within the British Athletics Classification system described by Pollock et al.¹² (a) Myofascial (b) Musculotendinous (c) Intratendinous

The goal of management of an acute muscle strain is to return the athlete to activity at the prior level of performance and with minimal risk for re-injury. This requires the underlying pathology and the changes it introduces (i.e. pain, swelling, weakness, reduced range of motion), be addressed. It also requires causative risk factors be addressed, with a previous muscle injury being the single largest risk factor for a future strain.

Acute management of muscle strains focuses on minimisation of pain and oedema, restoration of neuromuscular control at slow speeds and prevention of excessive scar formation while protecting the healing fibres from excessive lengthening. Recommended early techniques include: 1) ice and compression; 2) mobilisation and motion, within pain limits and while avoiding aggressive stretching techniques; and 3) gentle massage of the affected muscle peripheral to the lesion.¹⁵

Subsequent management allows for increased exercise intensity, neuromuscular training at faster speeds and larger amplitudes and the initiation of eccentric resistance training. The final stage of recovery progresses to high-speed neuromuscular training and eccentric resistance training in a lengthened position in preparation for return to sport.¹⁶

Contusion

Muscle contusion is the medical term for a muscle bruise, and refers to bleeding and subsequent haematoma formation within a muscle and its surrounding connective tissue sheaths. Contusions occur when a muscle is compressed against the underlying bone by a blunt, external force. The force can result from a direct blow or collision with an external object, such as an opposing player, teammate or piece of equipment. Thus, they are common in contact sports and sports involving the use of rapidly moving hard objects (e.g. sticks and balls).

The most common site for muscle contusion is the quadriceps, with contusions at this site being referred to as a 'corked thigh', 'charley horse' or 'dead leg'. Other common sites for contusion include the gastrocnemius and gluteal muscles; however, a contusion can occur in any muscle exposed to a blunt compressive force.

Muscle compression causes muscle fibre damage and rupture of microvessels. The released blood clots to form a haematoma that initiates an acute inflammatory reaction aimed at removing the haematoma and damaged tissue and initiating a repair response. The haematoma may form either within (i.e. intramuscular) or between (i.e. intermuscular) the fascial coverings of the muscle. Intramuscular contusions are more self-limiting as the rise in pressure within the confines of the fascial compartment reduces blood flow, but they are also generally more painful and more restrictive because muscle contraction results in further rises in compartment pressure and stimulation of nociceptors.

In contrast, intermuscular contusions are generally less painful as the fascial sheath is damaged which allows the haematoma to spread to relieve compartment pressure. Intermuscular haematomas are generally more evident externally as the blood is able to travel distally due to gravity and into the subcutaneous tissues resulting in a visible bruise.

Initial management of a muscle contusion involves assessing for presence of an acute compartment syndrome (discussed below), controlling bleeding to reduce secondary injury (i.e. muscle fibre hypoxia) and protecting the injured site from further injury. Techniques include compression and ice with the muscle placed in a pain-free stretched position to help stop the bleeding and reduce muscle spasm. Techniques such as heat, alcohol and

vigorous massage, which have the potential to increase blood flow and/or cause a rebleed, should be avoided in the acute and subacute stages. Subsequent management focuses on restoring muscle function via progressive stretching and strengthening. The use of protective equipment in the form of force dissipating and dispersing padding should be considered to prevent re-injury upon return to activity. However, the reduction in injury risk gained by using protective equipment will need to be weighed against any associated reduction in athlete mobility and performance.

Myositis ossificans

Myositis ossificans is a form of heterotrophic ossification and refers to the formation of bone within a muscle. It is an infrequent complication of a contusion injury wherein bone-forming osteoblasts invade the haematoma and begin to lay down bone. The cause remains unknown; however, risk factors include severe contusions that limit joint range of motion (e.g. thigh contusions that limit knee flexion to $<45^\circ$), repeat contusion injury and inappropriate initial management that causes a rebleed.

Myositis ossificans should be suspected in muscle contusions that do not resolve in the expected time. Particular signs and symptoms include initial improvements in range and pain followed by subsequent deterioration

and ongoing or reappearance of inflammatory symptoms (i.e. resting, morning and night pain). The bone grows from 2–4 weeks after injury at which time an area of calcification may be visible on X-ray (Fig. 3.7a) or ultrasound (Fig. 3.7b) and a firm lump may be felt. Once active bone formation ceases the area of calcification gets slowly reabsorbed. As myositis ossificans is self-limiting, management is typically conservative and consists of anti-inflammatory approaches and non-painful stretching and strengthening.

Acute compartment syndrome

Muscles in the extremities are surrounded by a strong, thick connective tissue called fascia. The fascia serves as an attachment site for muscle, aids in force transmission and forms non-distensible compartments that facilitate muscle-pump mediated venous return. Injury leading to swelling or bleeding into one of the fascial compartments (usually the flexor compartment of the forearm or anterior compartment of the lower leg) can result in the development of acute compartment syndrome.

Acute compartment syndrome occurs when interstitial pressure within a compartment exceeds perfusion pressure leading to the onset of ischaemia and, ultimately, cellular anoxia and death. It is characterised by pain that is out of proportion to the inciting injury, pain on passive stretch and at rest, paraesthesia and pulselessness (although the

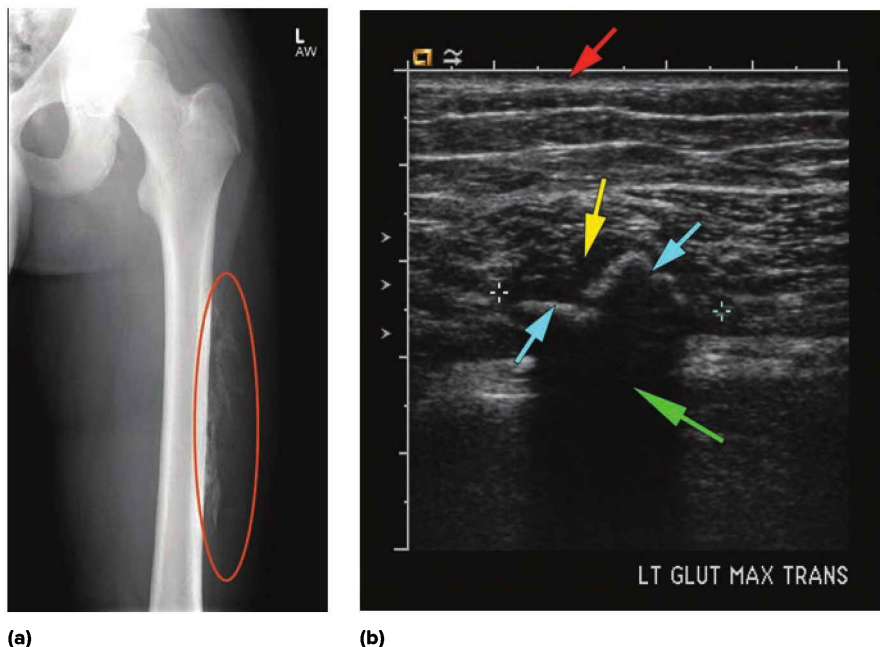


Figure 3.7 Myositis ossificans (a) X-ray appearance (b) Ultrasound appearance: yellow arrow points to haematoma within gluteal muscle; blue arrows to calcification within gluteal muscle; green arrow to acoustic shadowing, characteristic of calcification. Red arrow is skin and then subcutaneous fat, and below it are layers of gluteal muscles

latter is a late finding as the interstitial pressure needs to be high enough to occlude arterial flow). Acute compartment syndrome represents a medical emergency that may require urgent fasciotomy (i.e. release of the fascia surrounding the muscle compartment) in order to prevent permanent, irreversible damage.¹⁷ Hence, referral to an orthopaedic emergency specialist is recommended.

Cramp

Muscle cramps are sudden, painful, involuntary contractions characterised by repetitive firing of motor unit action potentials. When they occur during or immediately after exercise in healthy individuals with no underlying metabolic, neurological or endocrine pathology, they are referred to as 'exercise associated muscle cramps' (EAMC).¹⁸ EAMC are usually temporary (lasting <60 seconds), but are often incapacitating and tend to recur if activity at the same exercise intensity level is continued without an adequate recovery period. Thus, EAMC can significantly impair athletic performance. The most common sites for EAMC are in the calf and foot muscles, followed by the hamstring and quadriceps muscles.

EAMC were historically thought to be due to dehydration and/or electrolyte depletion, but this hypothesis has not been supported by scientific evidence. As EAMC most commonly occur towards the end of or after fatiguing exercise, or following a rapid increase in exercise intensity, fatigue appears to be a significant contributing factor. The current leading theory is that EAMC result from abnormal neuromuscular control at the spinal level in response to fatiguing exercise.¹⁸ This theory ascribes EAMC to increased excitatory and decreased inhibitory afferent inputs to motor neurons during fatigue. The net result is sustained motor neuron activity and the production of a cramp-inducing discharge.

Immediate treatment of EAMC aims to reduce motor neuron activity, with the most popular and effective technique being passive stretching. Passive stretching increases the Golgi tendon organ's inhibitory activity to reduce muscle electromyographic activity within 10–20 seconds and provide symptomatic relief. Ideally, passive tension should be maintained to the affected muscle for up to 20 minutes or until fasciculation ceases.¹⁹ If the EAMC were particularly intense or prolonged, icing while the muscle is on stretch may be considered to offset any potential delayed onset muscle soreness resulting from microscopic muscle fibre damage. Other potential methods of reducing motor neuron activity during EAMC include electrical stimulation of tendon afferents and antagonist contraction to induce reciprocal inhibition.

The incomplete knowledge of the aetiology of EAMC has limited the development of preventative strategies.

Fluid and electrolyte replacement has not been shown to be beneficial, consistent with the now debunked dehydration-electrolyte imbalance theory. In contrast, strategies aimed at modulating fatigability and altering neuromuscular control may be beneficial. Fatigability may be modified by improving generalised conditioning and endurance, and ensuring and maintaining adequate carbohydrate reserves. Neuromuscular control may be targeted by performing plyometric and eccentric exercises, which may elicit changes in muscle spindle and Golgi tendon organ firing to enhance efficiency and sensitivity of reflexive and descending pathways used for neuromuscular control. Regular stretching is also recommended as EAMC frequently occur when the muscle is in its shortened position, while massage and treatment of myofascial trigger points may improve muscle efficiency and reduce fatigability.

TENDON

Tear/rupture

Tendons connect muscle to bone and function to transmit muscle contractile forces necessary for motion. They consist of parallel collagen fibres that are more tightly packed than in ligaments, which endow tendons with the consummate ability to resist tensile loading. The tensile strength of tendon is so great that acute tear or rupture of a normal, healthy tendon is relatively rare, with forces more likely to cause an avulsion fracture or failure at the musculotendinous junction (i.e. muscle strain).

Tendons that do acutely fail usually have some form of underlying pathology (i.e. tendinosis). The pathology is usually asymptomatic, such that the tendon failure appears to occur without warning. As the presence of pathology increases with age, tendon ruptures most frequently occur in middle-aged and older athletes. The two most commonly ruptured tendons are the Achilles and supraspinatus tendons, with ruptures involving either a portion (partial rupture) or the full thickness (complete rupture) of the respective tendon (Fig. 3.8). Partial ruptures are characterised by the sudden onset of pain, localised tenderness and a loss of tendon function that is inversely related to the size of the tear. In contrast, complete ruptures are associated with total loss of tendon function and acute pain, but the pain often settles quickly due to concomitant damage of nociceptor afferents.

Diagnosis of tendon ruptures can be confirmed using imaging, with both ultrasonography and magnetic resonance imaging being useful at distinguishing between partial and complete rupture. Partial tendon ruptures may be managed conservatively using progressive rehabilitation; however, complete ruptures may be treated by surgical repair to restore tendon continuity and function, followed by rehabilitation or by rehabilitation alone.

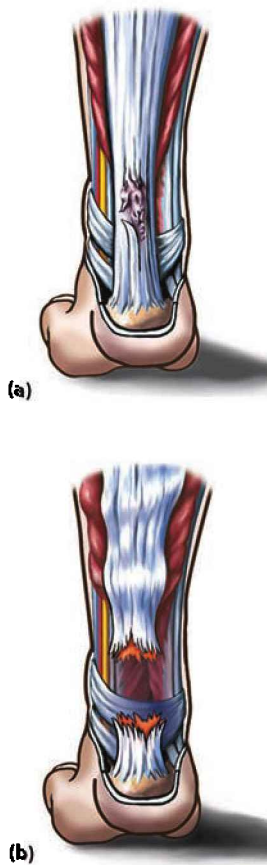


Figure 3.8 Tendon rupture (a) Partial (b) Complete

FASCIA

Tear/rupture

Fascia is a dense, regular connective tissue consisting of closely packed bundles of collagen fibres. It divides muscles into compartments and forms specific force-transmitting structures, such as the iliotibial band (ITB) and plantar fascia. Most injuries to fascia are associated with overuse, with prominent conditions being ITB syndrome (Chapter 37) and plantar fasciopathy (Chapter 43). Acute sprain or rupture of fascia is rare, but has been reported.

BURSA

Traumatic bursitis

Bursae are small, synovial membrane-lined sacs that are filled with an inner layer of viscous fluid. They are frequently found between bone and overlying connective tissues (such as tendon, muscle and skin) where they function to provide cushioning and facilitate movement by reducing friction. Most injuries to bursae are associated with overuse (Chapter 4), but occasionally a direct fall

onto a bursa may result in acute traumatic bursitis due to bleeding into the bursa. The management of acute haemorrhagic bursitis involves the application of ice and compression. Aspiration may be considered if the condition does not resolve, but corticosteroid injection is rarely indicated.

NERVE

Neuropraxia

Major acute nerve injuries are unusual in athletes. Specific nerves are susceptible to compression injury because of their subcutaneous location, including the ulnar nerve at the elbow and common peroneal nerve at the neck of the fibula. Similarly, some nerves are vulnerable to being co-injured with other tissues, such as axillary nerve injury during shoulder dislocation and radial nerve injury with humeral shaft fracture.

The most common acute nerve injury in athletes is a 'stinger' or 'burner'. A stinger or burner results from an overstretching injury to the brachial plexus that conveys the nerve supply to the arm. Overstretching can occur when the head is forcibly bent away from the shoulder while the shoulder is simultaneously depressed. This can occur when an athlete is driven into the playing surface during a high-contact sport, such as ice hockey, rugby, wrestling and various football codes.

Symptoms are a stinging or burning sensation that spreads from the shoulder to the hand, which may be associated with numbness in the sensory distribution of the involved nerves. The paraesthesia is often temporary, disappearing in minutes and usually within hours; however, injuries that are more serious can result in longer lasting symptoms involving the loss of both sensory and motor function due to blockage of nerve conduction. The later type of injury, known as 'neuropraxia', usually resolves spontaneously but slowly.

FAT PAD

Bruise/contusion

Fat pads consist of closely packed adipose cells surrounded by fibrous septa, which commonly divide the fat pad into separate compartments. The function of fat pads is not well established and may vary according to anatomical location. However, they appear to have roles in cushioning and facilitating joint lubrication. It is in their cushioning role that fat pads are susceptible to acute compression injury. In particular, the calcaneal fat pad under the heel and the infrapatellar fat pad located behind the patellar tendon can be acutely injured. The plantar fat pad can be injured during landing onto the heel from a height, while the infrapatellar fat pad can be injured by landing on the knees or pinching the fat pad between the femoral



condyles, proximal tibia and inferior pole of the patella during knee hyperextension. The infrapatellar fat pad is also often acutely injured when creating portals during knee arthroscopic surgery.

Fat pads are well vascularised and innervated. The blood supply contributes to haematoma formation (contusion) in response to acute injury, but also enables fat pad injuries to readily heal. The rich innervation of fat pads means that they can be a significant source of pain, with the infrapatellar fat pad being reported to be the most pain evoking structure in the knee.²⁰ The integrity and function of the calcaneal fat pad may be compromised if its organised compartments are disrupted; however, this is rare, with compression injuries sufficient to fracture the calcaneus not damaging the fibrous septa.²¹ Management

of fat pad injuries, beyond acute care techniques, includes externally padding using heel cups or knee pads and using tape to limit radial expansion of the calcaneal fat pad or unload the infrapatellar fat pad. These injuries can take considerable time to heal.

SKIN

Acute skin injuries are common in athletes, particularly those competing in contact sports and cycling. Open wounds may be caused by a scraping (abrasion), cutting (laceration) or piercing (puncture) force. Possible damage to underlying structures, such as tendons, muscles, blood vessels and nerves, should always be considered. The principles of treatment of all open wounds are shown in Table 3.4.

Table 3.4 Principles of treatment of all open wounds

Principle	Details
1. Stop any associated bleeding	Apply a pressure bandage directly to the injured part and elevate it. If the wound is open and clean, bring the wound edges together using adhesive strips or sutures. A contaminated wound should not be closed.
2. Prevent infection	Remove all dirt and contamination by simple irrigation. Extensively wash and scrub with antiseptic solution as required as soon as possible. If the wound is severely contaminated, prophylactic antibiotic therapy should be commenced (e.g. flucloxacillin, 500 mg orally four times a day). If anerobic organisms are suspected (e.g. wound inflicted by a bite), add an antibiotic such as metronidazole (400 mg orally three times a day).
3. Immobilisation (where needed)	This applies when the wound is over a constantly moving part, for example, the anterior aspect of the knee. Certain lacerations, such as pretibial lacerations, require particular care and strict immobilisation to encourage healing.
4. Check tetanus status	All contaminated wounds, especially penetrating wounds, have the potential to become infected with <i>Clostridium tetani</i> . Tetanus immunisation consists of a course of three injections over 6 months given during childhood. Further tetanus toxoid boosters should be given at 5 to 10 year intervals. In the case of a possible contaminated wound, a booster should be given if none has been administered within the previous 5 years.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

